# Unit - 3 Production

Production is very important concept of economics. It is the part of metro economics. It is depend on demand & demand forecasting. It means we produce a product according to the demand & after the estimation of demand. we can increase or decrease the production. It is the creation of utility mean by production, we mean the process by which someone utilise or convert the natural resources. There are 2 applicable terms in production, first one physical production in which we include bread, better, jam, milk and in the non-physical terms, we can include all type of services like doctor, engineer, transportation, banking & insurance. By production. Human being can use the particular product and often the consumption of product, they get satisfaction In simple words, we can say that on the basis of production, human can use the particular product & get more satisfaction

There are 3 types of utility.

(1) <u>Form Utility</u> :- When the existing matter is transformed on rearrange to make it more useful to satisfied human wants called the Form Utility.

By changing a particular product in their ~~organic~~ original form to new form to make it very useful, It include in the form utility.

The old one have lesser value than the new one.

(2) Place utility :- When the existing matter is transported to some another place according to their use is called place utility. In this type of ~~+~~ utility, product value, lesser but its transported to another places, their value would be higher to become more useful.

(3) Time utility :- In the time utility there is always a time log between production & the consuption of goods. Goods is useful when it reaches according to time when the time ~~lags~~ laps there is no value of the goods. ~~Apart~~ Apart of that we also includes labour & capital human wants & economic value.

for eg.-> When medicine is necessary for time being but not after that time is passed.

## Production Function :-

### Factors of Production:-

There are 4 factors of production.

(i) Labour as a factor of production
(ii) Land as a factor of production
(iii) Capital as a factor of production
(iv) ~~E~~nterprienure as a factor of production

### (i) Labour as a factor of production.

Labour means a type of human effort. This is very important element in production In labour there is a human effort which is useful while working. labour
which is directed to be produced goods & services. After the working labour

There are some important charatorstic of labour.

(i) It is the human effort.
(ii) Labour sales their labour not itself.
(iii) Highly perishable
(iv) Weak bargaining power
(v) All labour is not productive

## Factors Affecting Labour:-

(i) Climate factor
(ii) Organizational factor
(iii) Education & training
(iv) Motivating factor
(v) Working conditions.
(vi) Personal qualities.

(ii) Land as a factor of production:-

Land doesn't mean soil but it's including in the natural resources, the term land is free gift of nature, water is also gift of nature but we include land as a factor of production because when these natural resources comes under & ordership of an individual, it becomes more useful. There are some characteristics of land.

(i) Free gift of nature
(ii) Land has some original power.
(iii) ~~The~~ Doesn't produce itself
(iv) Reward is rent.
(v) Inelastic supply.

(iii) Capital as factor of production:-

It is an important aspect of production Capital doesn't mean a simple money but capital means a perticular wealth. Capital is the production. Capital generate by men-made factors of production like tools, machinery plant equipments.

It is the part of wealth of an individual. There are some important differences between wealth & capital. In wealth we include goods & q human quality. which is useful in production. but In the capital we include some goods or services which produce money.

There are some characterstic of capital.

(i) Capital is not income.
(ii) It is not a natural resources.
(iii) Capital is the men-made formate.
(iv) It is available in different forms.

Classification of capital:-

(A) Real (equipement) & financial (eg money form) capital
(B) Private (personal) & social (Charity/ Govt property) capital
(C) Fixed (Saving) & floating (Share market) capital
(D) Tangabale (Sensable/ Cash) & Intangable (Brand image) Promotion capital value
(E) Indian & Foriegn capital.

(iv) Enterprienure as a factor of production.

One of the most important factor in today's world. Enterprenure is a person who generate new idea & start the particular idea. Enterprenure is the process by we make money & taking risk for future. It is a person or a group of person which is also controlled by government. There is some important function -

(A) Coordinating function.
(B) State of technology.
(C) Quality of product.
(D) Concept of Idea.
(E) Innovation factor.
(F) Risk bearing function.
(G) Some time meet supervision & direction.

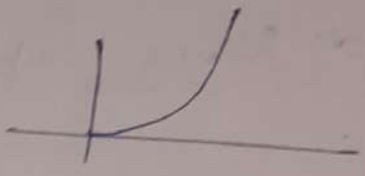
## Product function in Long run :-

It is important topic of economics. In Product function there are some three important stages -

(1) **Increasing return to scale** :- all factors will variable

In this stage the total output increase more then proportionately. is called
The main reason behind that is

(a) Motivation of work
(b) Good Hiring / requitment
(c) Good working conditions
(d) Skilled labour

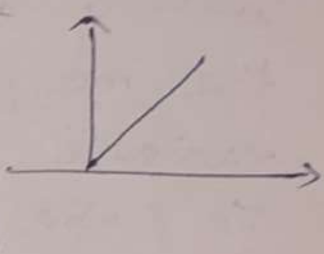


$2x \to 3x$

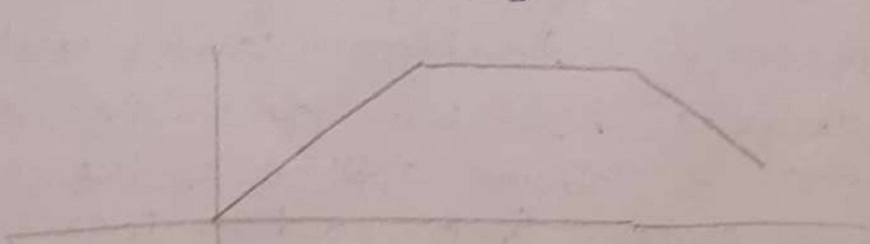
(2) **Constant return to scale** :-

In this stage the total output increase in the same proportion to that of factor ~~proportion~~ known as constant return to scale. The reason behind that is

(a) Managerial efficiency at same level
(b) No technological changes is needed
(c) Balanced expension of input
(d) Stable organizational structure.
(e) Perfect input available in market
(f) Stable external conditions.



$x \to x$



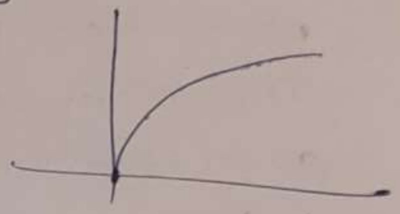
## (3) Dimnishing return to scale:-

It is also known as decreasing return to scale. In this stage the total output increase less then proportionality known as decreasing return to scale.

The main reason behind that is.

(a) limited natural resources

(b) Managerial complexity



(c) Technological limitation

(d) Rising input cost

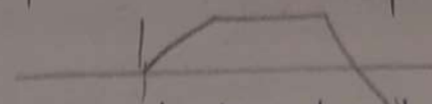
(not in negative) $x \rightarrow 0.5x$

## Production function in Short Run:- Land fixed Enterprenure

Production function in short Run is imp. point in the economist point of view. As economic analysis as we increase the quantity of one input which is combined with a fixed quantity of the other input the marginal physical productivity of the variable input must be decline.

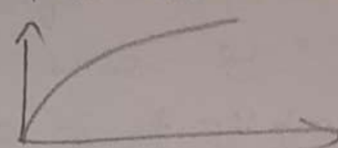
In simple words, we can say that in production function short Run as equal to input & output but all the stages are not same. In this we also include some input which is related with other fixed input will give a state of technology causes output increase or decrease. At last an increase in the capital & labour applied in the basis of land cause unless & proportion increase in the amount the product. ~~unless it~~

Unless it happens ~~imf~~ with an improvement in agriculture.



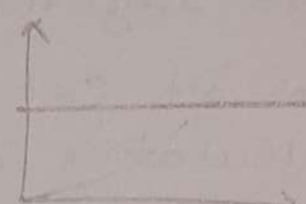
There are some three important stages:-

(1) <u>Increasing return to factor:-</u>
At this stage the total output increase is more than the proportionate increase of input. That means the total output increase more then porportionatly input.
The main reason behind that is
(a) good working condition, with no max profit & loss situation.
(b) Good hiring of the labour.
(c) Balanced worthies.



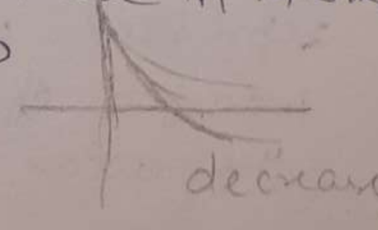
$2x \rightarrow 3x$

(2) <u>Dimnishing return to factor:-</u> In this stage where the output increase proportionality less then increase in variable input. The main reason behind that is
(a) Short term project
(b) Unskilled labour



$2x \rightarrow 2x$

(3) <u>Law of Negative Returns:-</u> In this stage the total output decrease rather than increase. The main reason behind that is
(a) limited resources available in market.
(b) labour capital ratio
(c) Short duration.




$2x \rightarrow x$

* **Cost Concept** :- Cost is an important Concept of economics. The cost function of the firm depends upon the physical production & it is the most important factor governing the supply of the product.

Cost analysis refers to the study of behaviour of cost in relation to one ~~are~~ or more production creteria the main important thing is that original cost always return in the account book of the firm & market cost is differ from original cost.

There are some important type of Cost -

- **Implicit & Explicit Cost** :-

Implicit cost also known as economic cost Implicit cost are those cost that the firm will make economic profit. When total revanue is implicit & explicit cost taken togethor.

Implicit Cost is ~~so~~ helpful for reporting, calculating controlling & useful for decision making.

ex Adjust the money ~~isp~~ in the another form.

while explicit cost are those cost which is known as accounting cost this is the actual cost which we use for payment. This cost represent ~~which we use for~~ pay actual transfer of money that is recorded into the account books. Accounting cost will take into account only the payment & charges made by the enterpreneure to the supplier of such resources.

Ex Actual pay the money.

(2) ~~Opposit~~

(2) Opportunity Cost :-

Opportunity cost are those cost which we earn after the particular opportunity. It is those type of cost in which we choose the best alternative has been selected. It is cost of displayed alternative in simple words we can say that opportunity cost is the best alternative could be produce in the place of some factor or by equalent to the other factor. This is the cost of the same amount of money.

(3) Private cost & Social Cost :-

Private cost is those cost which is our personal income. We personally use this cost for purchasing a particular product while social cost is those cost which is the firm wear this cost with the reference of government & also fully paid by government. This called Social cost.

(4) Shut Down Cost & abandonment Cost :-

It refers to those cost which would be inocurred in the event of a temperary session of a business activity. And it could be saved if operations are ~~cou~~ continue..
while abandonment cost is those cost which is the cost of retiring a fixed asset from its used

(5) Sunk cost :- Sunk cost are those cost that (डूबी हुई कीमत) has already been incurd and cannot be recovered. Regardness of any future decision.

(6) Direct cost & Inderect Cost :-

Direct cost are explicit cost while indirect cost are implicit cost.

## * Cost function in Short Run :-

(1) Total Fixed Cost :- TFC is the fixed cost which remains same irrespective of the level of output. In this we include the fixed expenses which is incurred by the firm. One the fixed factor of product. (like market Price)

(2) ~~Total~~ Total variable cost :- TVC is the cost which (like labour cost) varies with the production. They are the cost that are incurred variable factors the amount of which can be alter in the short run.

(3) Total Cost :- ~~TFC + TC~~

$$\boxed{TFC + TVC = TC}$$

(4) Avarage fixed Cost :- AFC is the those cost which is obtained by deviding TFC by the level of output.

$$AFC = \frac{TFC}{\text{level of o/p}}.$$

(5) Avarage variable cost :- AVC cost is total variable cost devided by the quantity of output produced.

(6) Marginal Cost :- MC is the addition to the total cost incurred on the production of additional output.

* Cost function in Long Run :-

(1) Long Run avg Cost :- LAC is the avg cost per unit of output when all the factors of production is important for the producti then the ~~fixe~~ best proportion of fixed & variable factors related with the long run.

(2) Long Run marginal Cost :- LMC can be defined as the additional cost producing an extra unit of output in the long run.

## Determining of Cost Function

(1) Level of output
(2) Size of the plant
(3) Price of input
~~Marginal efficien~~
(4) Managerial efficiency
(5) State of technology.

Q Explain Extent of a market & factor affecting it.

# Assignment - 3

Q1 Explain production function. Explain with the help of diagram.

Ans

## Product Function

A Production function shows the relationship between inputs (factors of production) & output.
It tells us how much output a firm can produce using different combinations of inputs.

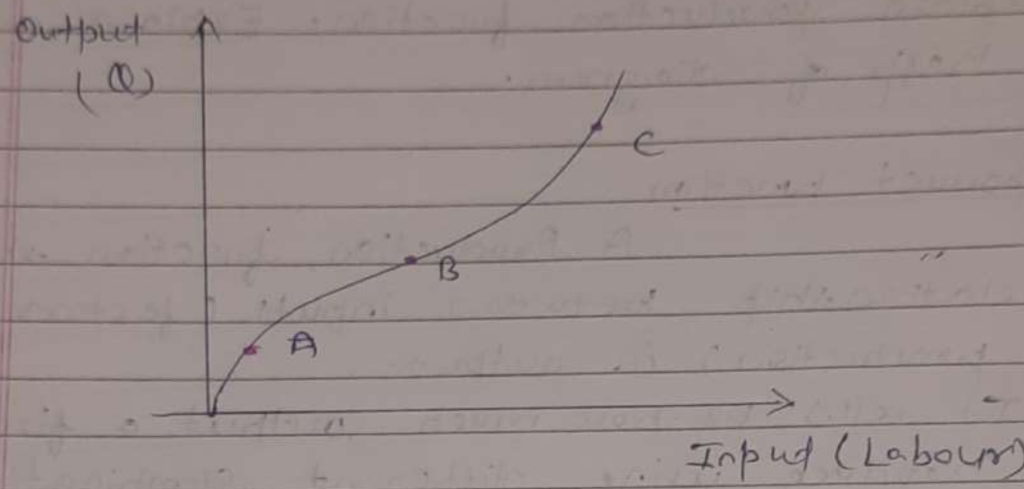
$$Q = f(K, L, R, E)$$

where factors of production -
K = capital → machinery, equipment, tools.
L = labour → human effort.
land → natural resources.
E = Entrepreneurial ability → mangement & risk - taking.
R = Raw materials.
& Technology → improvements that increase productivity.

## Production function in Short Run

In short run, one input is variable & other remains fixed.
Labour (L) = variable
Capital (K) = fixed

function → $Q = f(L, \bar{K})$




Here Different stages -

Stage (A) Increasing Returns (MP rising)
means total output increase is more then the proportionate increase input.
$(x \rightarrow 3x)$

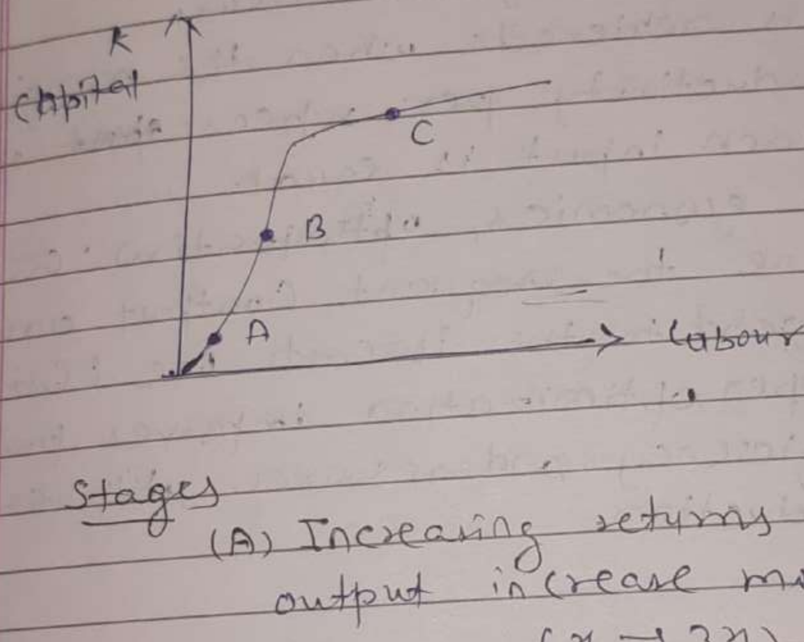
(B) Diminishing returns (MP falling)
means output increase proportionality less then increase in variable input.
economically efficient stage.
$(x \rightarrow x)$

(C) Negative Returns -
means the total output decrease rather then increase.
$(x \rightarrow x/3$ or $-x)$

Production function in long term:-
In long run all inputs are variable & the firm can chouse the best combination of labour & capital.

Function ⇒ $Q = f(L, K)$

Long run uses.
· Achieving least cost input combination.
· Planning scale of production
· Making investment decision..




Stages
(A) Increasing returns to scale →
output increase more than inputs
$(x \to 3x)$

(B) ~~Increasing~~ Constant returns to scale →
output increases equally to input.

(C) Decreasing returns to scale →
output increase less the inputs.

Q2 What do you understand by Production Optimization? a

Ans - Production optimization means using inputs in the best possible way to achieve maximum output or maximum cost.

- It helps firms decide the most efficient combination of labour, capital, and raw materials.
- The goal is to produce the desired output at the lowest possible cost without wasting resources.
- It is achieved when the marginal productivity per rupees ~~spent~~ spent on each input is equal.
- In economics, optimization occurs where the isoquant (output curves) is tangent to the isocost line (cost line)
- Proper optimization improves profit, efficiency and resource utilization in production.

Q3 Explain "Least cost Combination of input".

Ans The least cost combination means the firm uses inputs (labor, capital, material) in such a way the output is produced at the lowest possible cost.

It occurs where

Marginal Rate of Technical Substititution (MRTS)

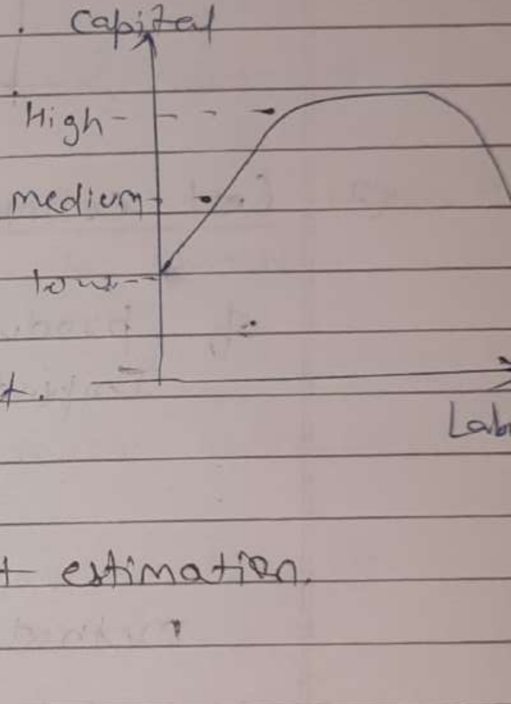
$$MRTS = \frac{\text{Price of Labor}}{\text{Price of Capital}}$$

Practical meaning -
- Firm chooses the input mix where the isoquant (desired output level) touches the isocost line (budget line)
- At this point, the firm is efficient & minimizes production costs.

Q4 What is isoquants?

Ans An isoquant is a curve that shows all possible combinations of inputs (Labor & capital) that produce the same level of output.

Characterstic -
- Downward sloping
- Convex to the origin
- Never intersect
- Higher isoquant = higher output.




Q5 Write a short note on -
(1) Cost curve (2) Cost estimation.

Ans.

(1) Cost Curve Cost curve shows the relationship between output & costs incurre by a firm.

Types of cost curve -
(a) T C (Total Cost) → Sum of fixed and variable cost

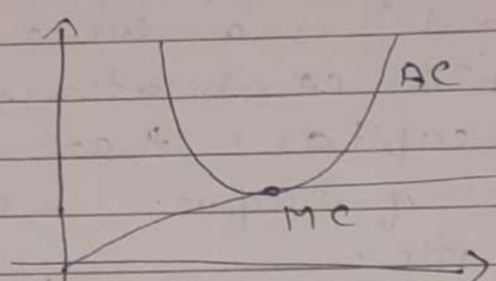
(b) TVC (Total variable cost) - which varies with output.

c) TFC (Total fixed cost) - constant with output.

d) AC (Average cost) - cost per unit.

e) MC (Marginal cost) cost of producing one more unit.

Shape = AC → U shape
MC cuts AC at its minimum pn.




(2) <u>Cost estimation</u>:- Cost estimation is a process of predicting future costs of production.

Importance - Helps in budgeting
- Assists in pricing decisions.
- Helps management judge profitability.

Methods -

Statical analysis (regression)
Engineering methods
Account analysis.

Q6 Explain Economies & Diseconomies of Scale scale along with its types.

Ans Economies of scale refers to the cost advantages a firm enjoys when it increase the average cost (AC) per unit decrease. Because fixed costs get spread over more units, & the firm becomes more efficient.

Types of economics of scale -

(1) Internal Economics of Scale :-
These arise with the firm, irrespective of the industry.

(a) Technical Economics :-
- Lower cost due to use of advanced machinery & technology.
- Example :- A large factory uses automation to produce goods faster & cheaper.

(b) Managerial Economies -
- Hiring specialized managers increases efficiency.
- Example :- Separate HR, marketing & production managers.

(c) Financial Economies -
- Large firms can borrow at lower interest rates due to better creditworthiness.

(d) Marketing Economies -
Bulk buying of raw materials at discount large-scale advertising spreads the cost

(f) Risk-bearing economies-
- Large firms can diversify products la markets, reducing business risk

(2) External Economies of Scale:-
These occurs outside the firms, due to industry growth.
(a) Economies of Concentration.-
- When many firms cluster in one area, they get access to skilled labor, suppliers and infrastructure.
- Example: Silicon Valley for tech firms
(b) Economies of information-
- Industry associations share R&D & market data.
(c) Economies of Disintegration.-
- Different firms specilize in different stages of production; lowering cost for all.

Diseconomies of Scale -

Diseconomies of scale occurs when a firm becomes too large, causing its average cost (AC) per unit to increase. Because inefficiencies start appearing at very large scales.

# Types of Diseconomies of Scale

## (1) Internal Diseconomies of scale

(a) Managerial Diseconomies -
- Communication becomes difficult, slowing decision-making.
- Too many employees → lack of coordination.

(b) Technical Diseconomies -
- Overuse of or underutilization of machinery due to large-scale operations

(c) Financial Diseconomies -
- Large firms may take excessive loans → higher risk → higher interest costs.

(d) Labor Diseconomies
- Workers may feel aliented in huge companies → lower productivity

## (2) External Diseconomies of Scale

(a) Higher Input Prices -
- When many firms compete for the same resources, prices go up.

(b) Congestion Diseconomies -
- Overcrowding of industrial ~~series~~ areas → High transportation delays & costs.